

Engineering Portfolio

Nathan Malyszek

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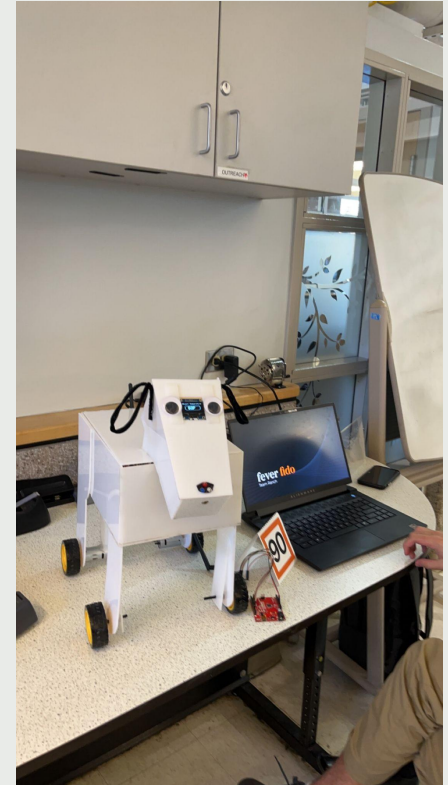
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Freshman Year Engineering Project

Fever Fido

Mission Objective: Provide an easy way for people to take their temperature in hospital settings by using a non invasive and approachable device.

- Touchless thermometer in nose
- LEDs in nose that light up depending on measured temperature
- LCD screen at top of head to show exact temperature
- Powered wheels on legs allowing for it to be controlled through laptop



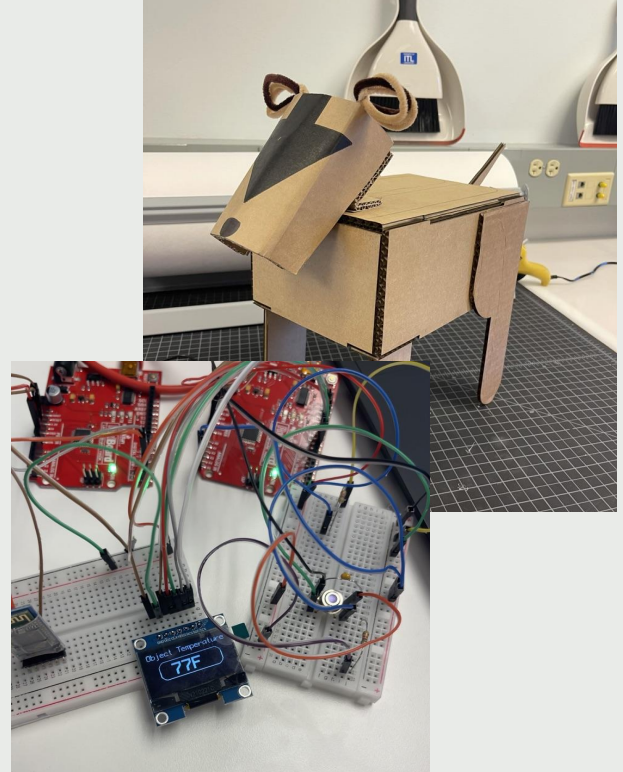
Freshman Year Engineering Project

Contributions

- Wiring for thermometer, LEDs, and computer
- Arduino code for temperature reading
- Laser cutting for body components

Skills Learned

- Prototyping
- Circuit design
- Arduino IDE and simple software
- Human factors design

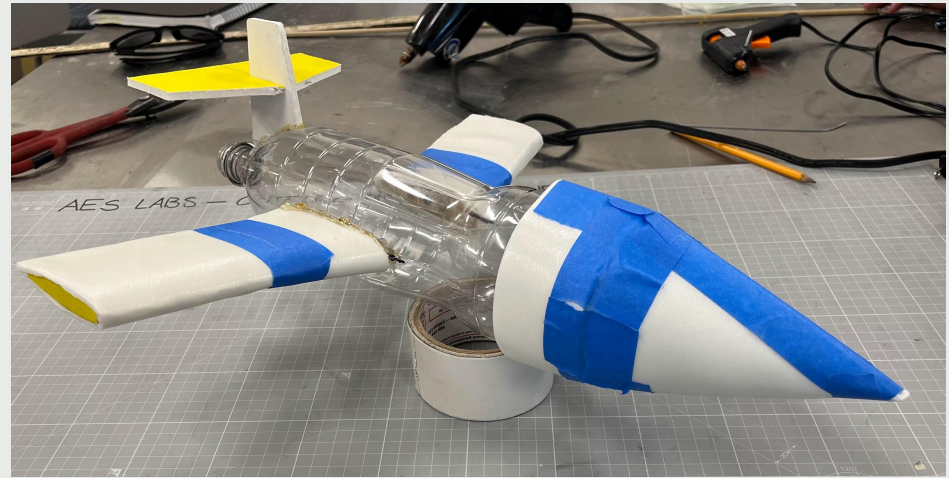


Sophomore Year Engineering Project

Boost-Glide Water Rocket

Mission Objective: Develop a small boost glide rocket that can travel 120 meters in the horizontal direction.

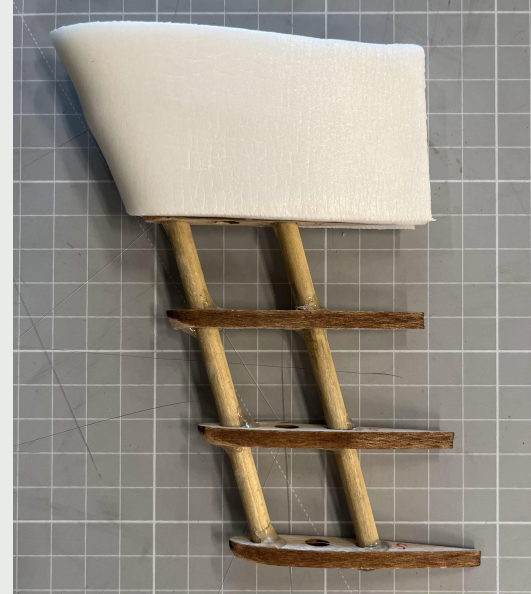
- 1.25L bottle
- Foam/balsa wings
- Trimmable horizontal stabilizer



Sophomore Year Engineering Project

Contributions

- Wing design and fabrication
- CAD design
- Nose cone
- Thrust testing



Sophomore Year Engineering Project

Results

Left: 27.16 meter boost

Right: 33.70 meter glide

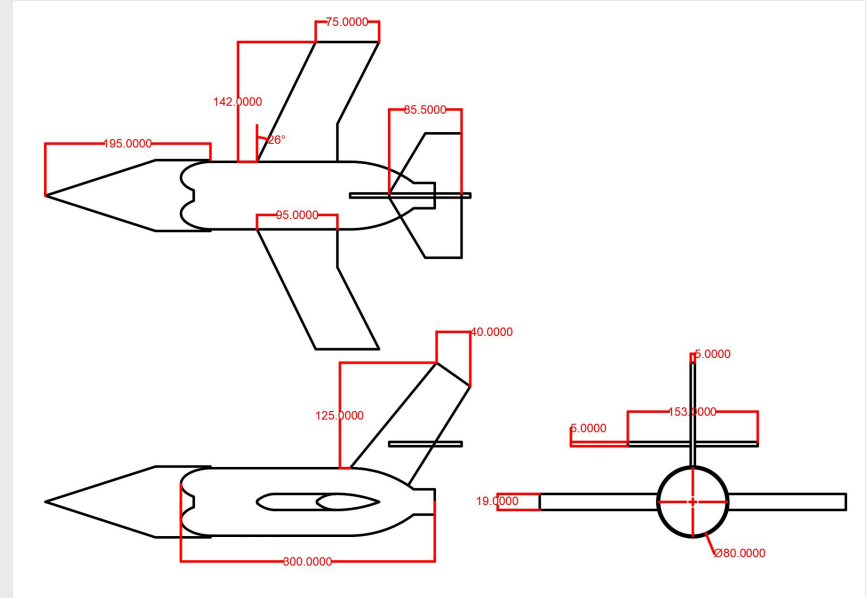
One of the best groups for
combined boost + glide
performance!



Sophomore Year Engineering Project

Skills Learned

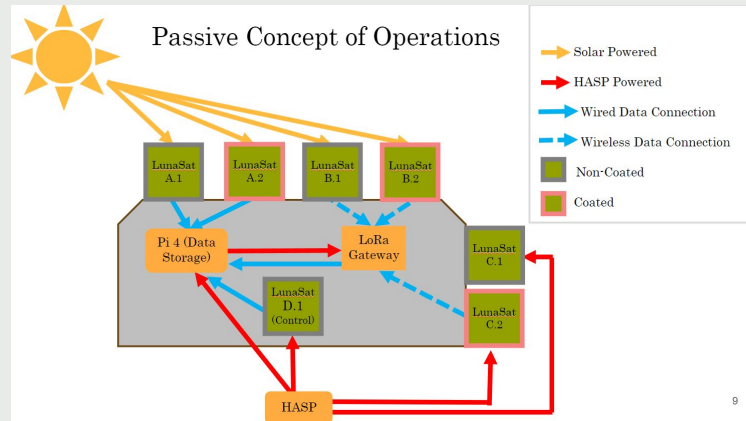
- CAD software (primarily AutoCAD + solidworks)
- Prototyping
- More laser cutting experience
- 3d printing
- PDR presentation
- Testing
- Aerodynamic and structural design



High Altitude Student Payload (HASP)

Mission Objective:

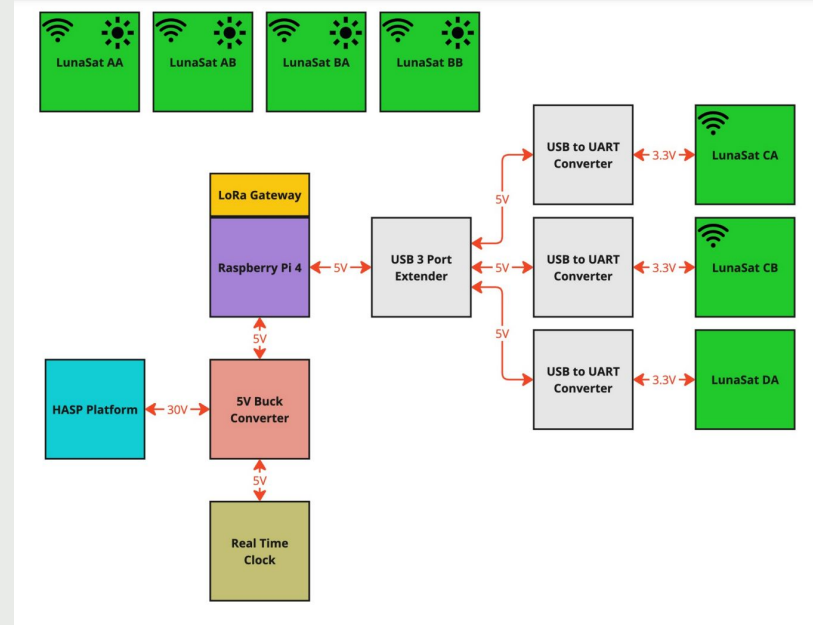
Develop a payload for NASA Colorado Space Grant that can fly on the HASP platform, testing LunaSats in a space-like environment. Payload needs to interface with platform connections for data and power, allowing for it to be monitored throughout the mission.



High Altitude Student Payload (HASP)

Contributions

- Initially science subteam lead
 - Designed entire experiment
 - Multiple testing blocks for redundancy
 - Different independent variables for each block
- Transitioned to avionics team lead as design progressed
 - Designed and tested power and data circuits
 - Integrated all avionics components into payload housing
 - Soldered, crimped, and built all connections/interfaces



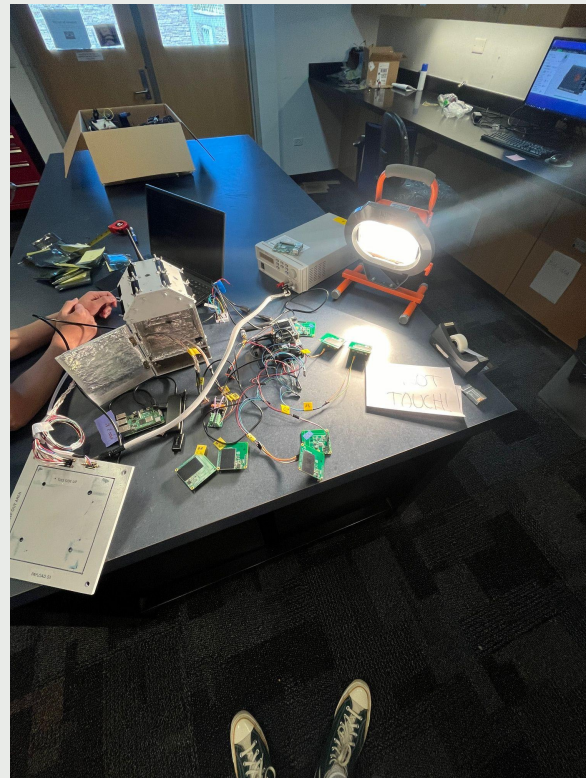
High Altitude Student Payload (HASP)

TVAC Testing

- Attended TVAC testing at the Columbia Scientific Balloon Facility to be certified for flight

Ground Station Monitoring

- During flight, I monitored and recorded downlink during flight. I also sent commands to the payload when needed



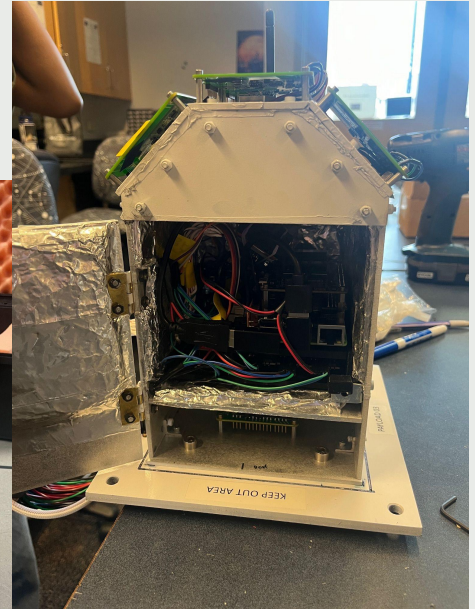
High Altitude Student Payload (HASP)

Results

- Successful flight in September 2024
- Payload remained functional throughout entire flight
- Electrical system survived delivery, flight, impact, and return

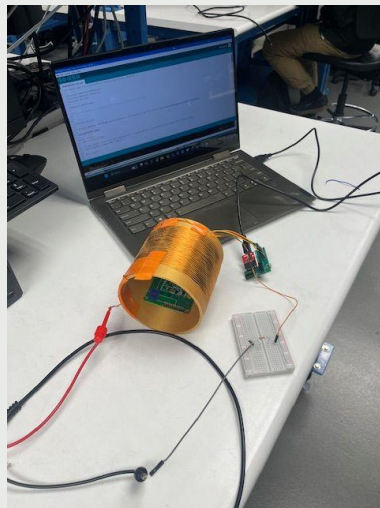
Skills Learned

- Circuit design and testing
- Prototyping
- Data systems
- Design reviews
- Full product development cycle
- Team leadership



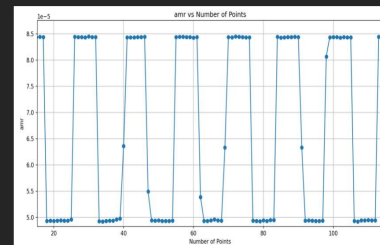
GLEE Science Lead

While working on HASP, I was also the GLEE (Great Lunar Expedition for Everyone) science team lead, I was in charge of testing the LunaSat technology as it develops. I managed work allocation for 4 team employees and set goals for the team.

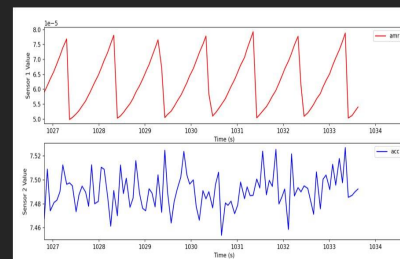


Magnetic Field Testing and Outputs

Square input 1Hz, 10Vpp, 5V offset:



Ramp input 1Hz, 10Vpp, 5V offset:

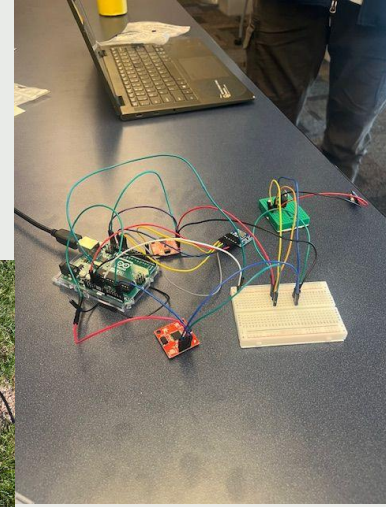


GLEE Science Lead

Daily work primarily consisted of testing sensors onboard the LunaSat, which is a small pcb that is designed to be distributed across the Lunar surface in the hundreds. These LunaSats will operate for 1 Lunar day, sending back data wirelessly to a central computer.

Skills Learned

- Team leadership
- Writing test plans
- Effective communication
- Quality control
- Documentation
- Data processing

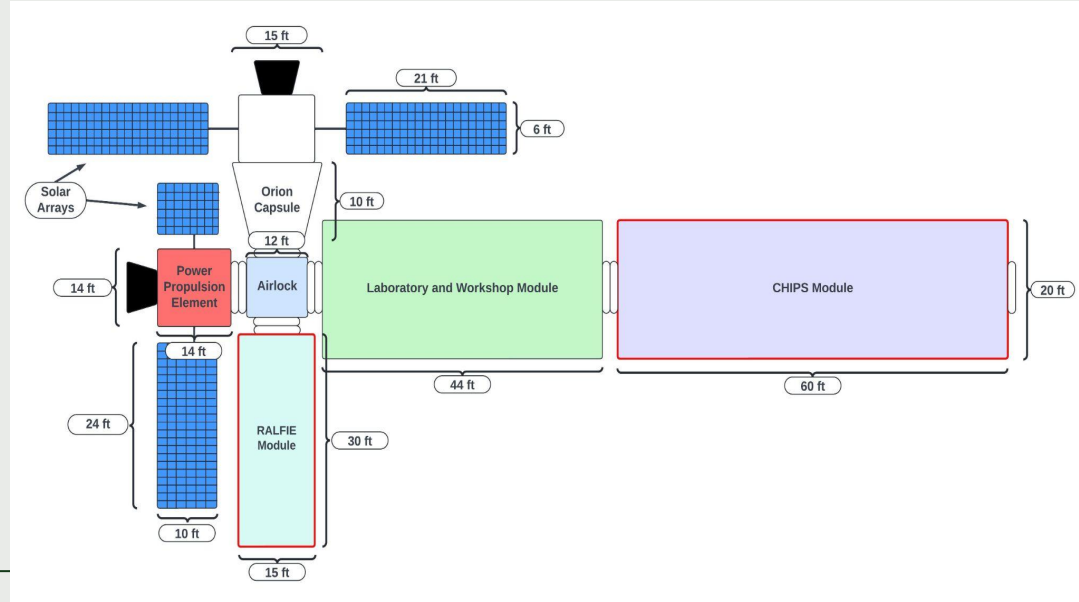


Accelerometer Sensor Field Tests

Human Spaceflight Project

For my Human Spaceflight final project, taught by former astronaut Jim Voss, I was tasked with designing the habitation module for 4 astronauts on a trip to Mars. I needed to determine everything that was required to sustain the astronauts on their round trip, ensuring that they were happy and healthy.

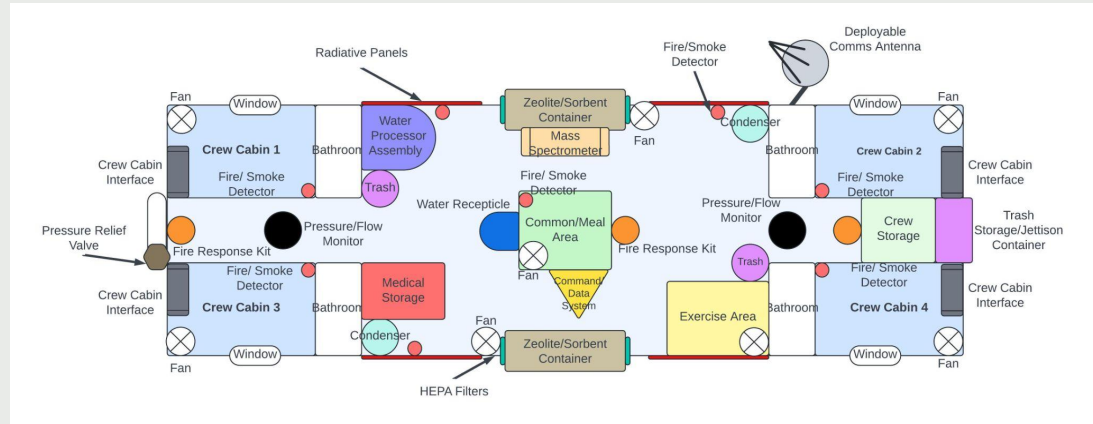
[Full Project Report Here](#)



Human Spaceflight Project

Lessons Learned

- Environmental Control and Life Support system design
- Mission design
- Human factors design
- Bioastronautics



Crew Habitation Module

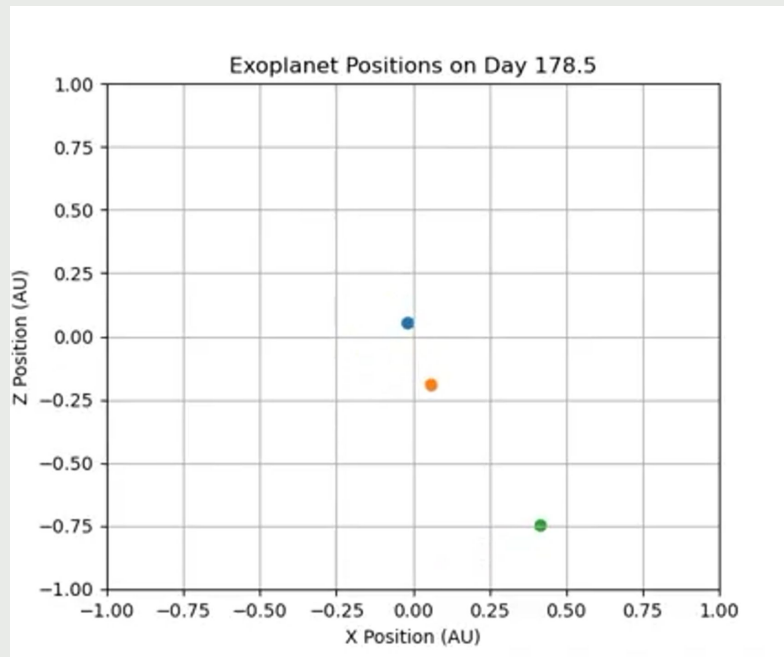
N Body System Simulation Project

This past summer, to improve my coding skills, I took an astronomy course that was based in Python. For my final project, I built a Python simulation that plotted the orbits of planets with various initial conditions.

Skills Learned

- Coding in Python
- Data processing
- Python libraries

[Code Here](#)

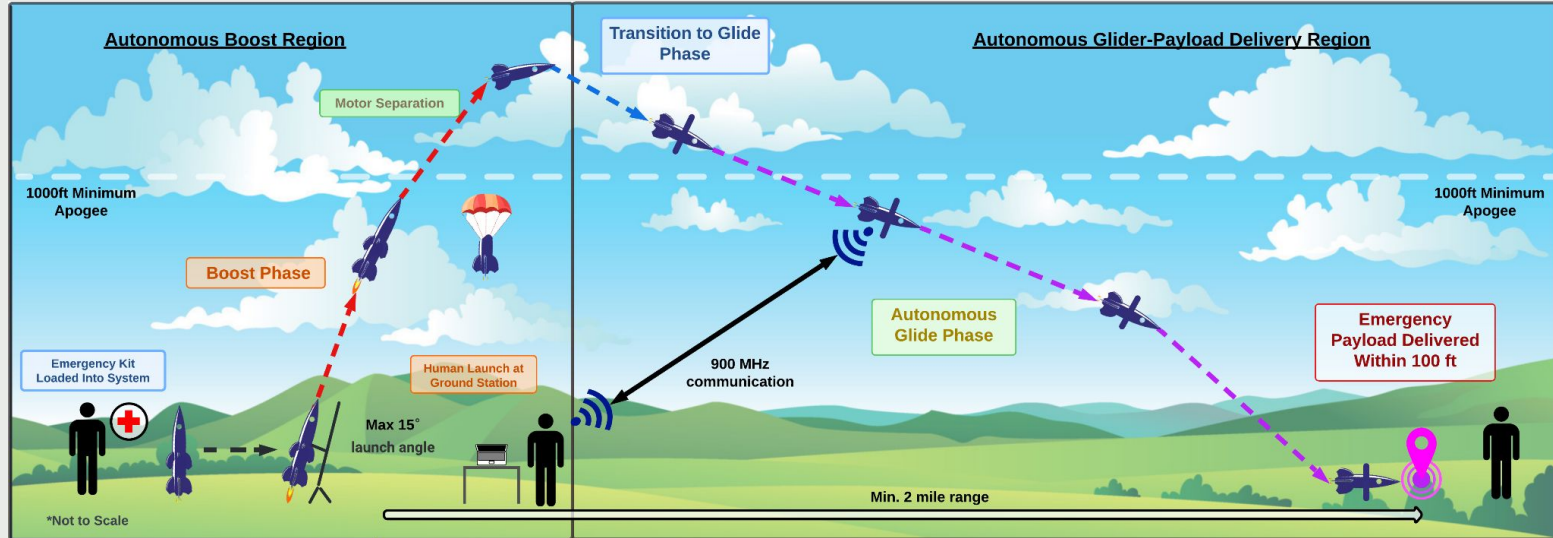


Three Body Simulation

Senior Project

Mission Objective:

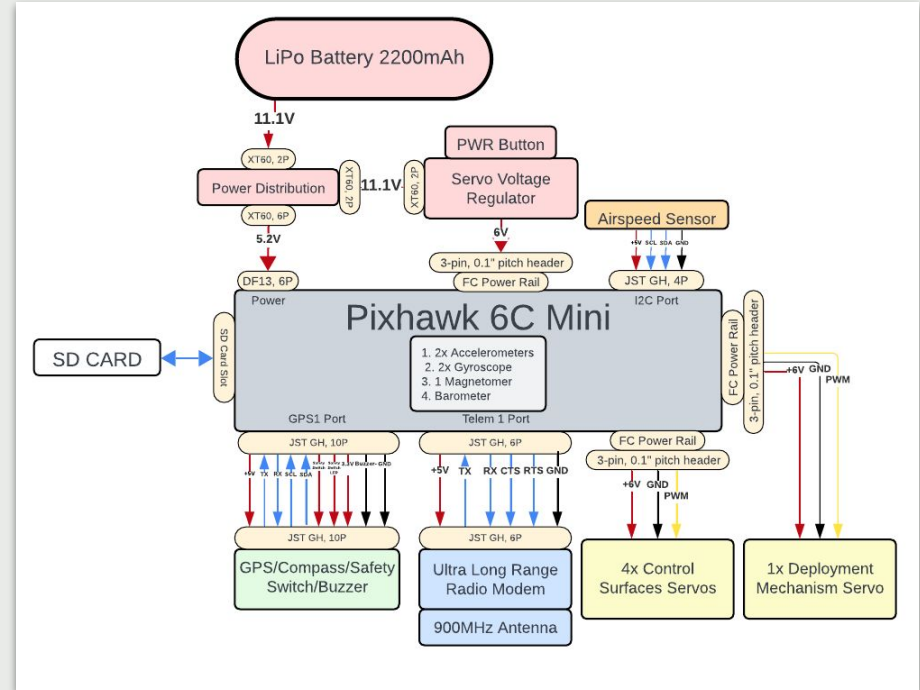
Accurately deliver an emergency rescue kit via rocket-glider to a stranded individual in a remote, hard-to-reach location. Dedicated to providing survival items in a time sensitive situation.



Senior Project

Contributions:

- Developed and tested avionics system
- Picked components needed to complete mission
- Mounting and harnessing
- Creating mission profile software for flight computer
- Attending and presenting in design reviews
- Developed and tested wing deployment system



Avionics System FBD

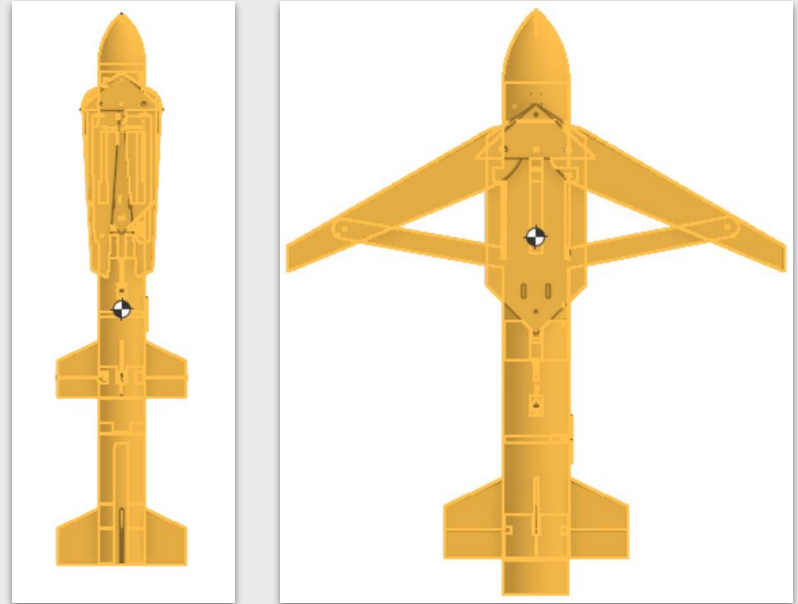
Senior Project

Fall 2024

- Passed both PDR and CDR presentations in Fall 2024
- Moved forward to manufacture and testing in spring

Skills Learned

- Effective presentation and communication
- Ardupilot software
- Working on large teams
- Trade studies
- Amateur rocketry



Current CAD Design for Rocket Glider

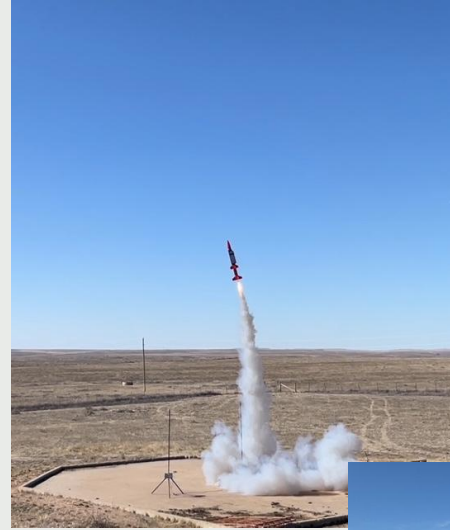
Senior Project

Early Spring 2025

- Created multiple prototypes for testing and analysis
- Developed more in depth models to verify stability

Boost Test

- Manufactured first boost test in less than 3 months
- Learned valuable lessons about stability, structural weaknesses, and mounting options



First Boost Test with "Big Red"

Senior Project

Late Spring 2025

- Manufactured full scale system for final complete test
- Continued to present work at design and test reviews
- Received award for "Best Simulation and Modeling"
- Final test cancelled due to dry conditions and fire hazard

Skills Learned

- Manufacturing and machining techniques
- Testing safety
- Mechanical design
- Prototyping and rapid development

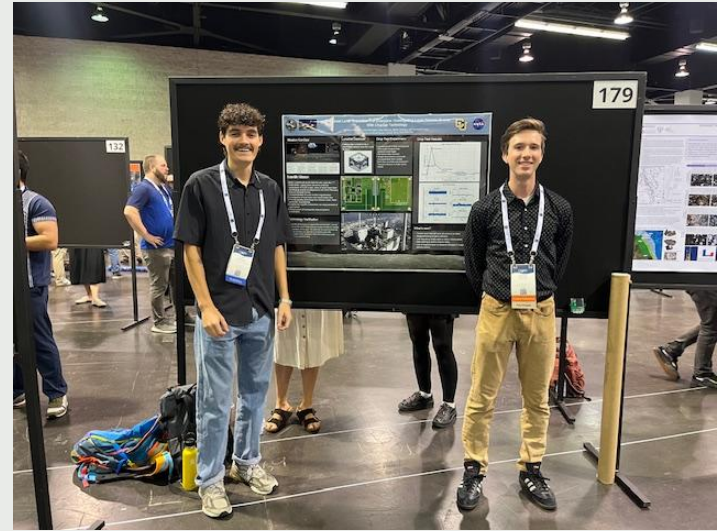


Final Product

Conference Presentations

In September 2024, I presented a poster in the undergraduate research section at the Geological Society of America (GSA) in Anaheim, California. My poster covered my work at NASA Colorado Space Grant Consortium regarding moonquake detection using a network of accelerometers on the Moon.

At this conference, I improved my networking skills and got to experience the academic side of planetary science for the first time!



Conference Presentations

In March 2025, I was also able to attend the Lunar and Planetary Scientific Conference (LPSC) and continued to represent the GLEE mission while meeting industry professionals.

This was a great opportunity to connect with scientists and engineers, where we pitched our mission and made great contacts in our mission to get a ride to the Moon!

